

CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Spring 2026

Lecture 12

1 Diagnostics of Multivariate Linear Regression Model

- List of Assumptions
- Residuals, Leverage, and Studentized Residuals
- Examples:
 - No Outliers: $E[X_i^4] < \infty$ and $E[Y_i^4] < \infty$
 - Large Residual, Small Leverage
 - Large Residual, Large Leverage
 - Small Residual, Large Leverage

2 Weighted Least Squares (WLS)

- Infeasible WLS: Known Heteroskedasticity
- Feasible WLS: Heteroskedasticity of Known Function Form
- Example

3 Matching Methods

- Example
- Exact Matching

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Assumptions of Multivariate Linear Regression Model

Let's recall the assumptions of a Multivariate Linear Regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

where

- ① *exogeneity* assumption: $E[u_i | X_{i1}, X_{i2}, \dots, X_{ik}] = 0$
- ② $(X_{i1}, X_{i2}, \dots, X_{ik}, Y_i)$ is i.i.d. sample drawn from a joint distribution
- ③ $E[X_{i1}^4], E[X_{i2}^4], \dots, E[X_{ik}^4]$ and $E[Y_i^4]$ are nonzero and finite
- ④ there is no perfect *multicollinearity*: none of the regressors is a linear function of the other regressors

and focus on the 3rd assumption and consider examples.

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Assumptions of Multivariate Linear Regression Model

Let's now also assume homoskedasticity with $\text{Var}(u_i|\mathbf{X}) = \sigma^2$ and define

- *Residuals*: $\hat{\mathbf{u}} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}$, where $\hat{\boldsymbol{\beta}}$ denotes the OLS estimator
- *Leverage*: $h_{ii} = [\mathbf{H}]_{ii}$ where $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is the *hat matrix*
- *Studentized Residuals*: $t_i = \frac{\hat{u}_i}{\hat{\sigma} \sqrt{1-h_{ii}}}$

Then

- 1 $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$
- 2 $h_{ii} = \frac{\partial \hat{Y}_i}{\partial Y_i}$
- 3 $\mathbf{H}^2 = \mathbf{H}$ and thus $[\mathbf{H}^2]_{ii} = [\mathbf{H}]_{ii}$ and $0 \leq h_{ii} \leq 1$
- 4 *Cook's Distance*: $D_i = \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})}{(k+1)\hat{\sigma}^2} \approx \frac{t_i^2}{k+1} \frac{h_{ii}}{1-h_{ii}}$,

where $\hat{\boldsymbol{\beta}}_{(i)}$ is the OLS estimate obtained when the i -th observation is excluded.

- 5 $D^2(X_i) \doteq (X_i - \bar{X})^\top (\mathbf{X}^\top \mathbf{X})^{-1} (X_i - \bar{X}) = (n-1)(h_{ii} - \frac{1}{n})$
is *Mahalanobis distance*

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Example: $E[X_i^4] < \infty$ and $E[Y_i^4] < \infty$

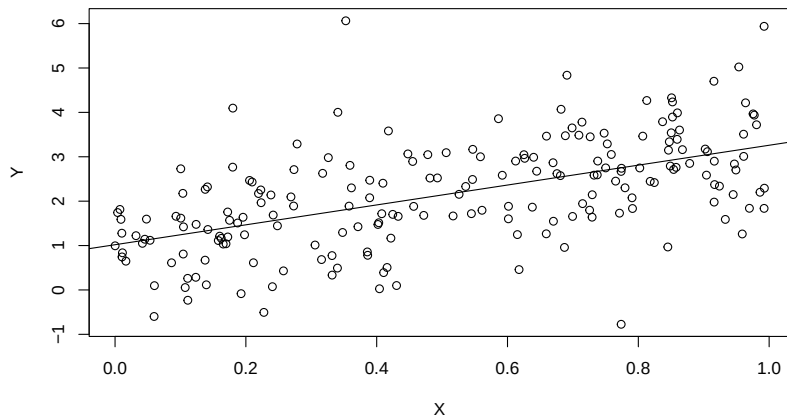
Example:

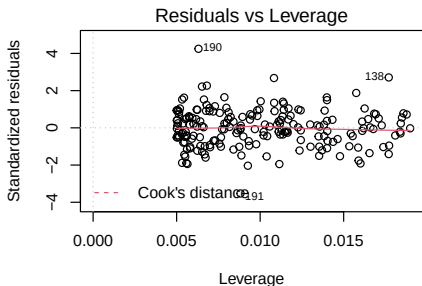
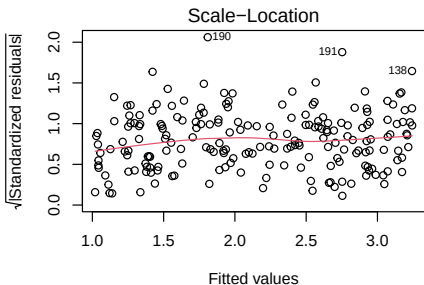
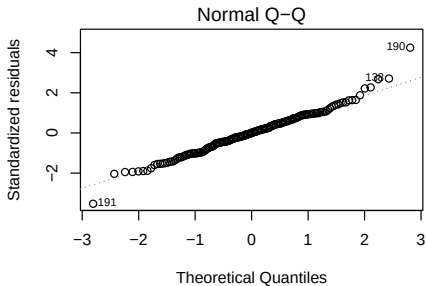
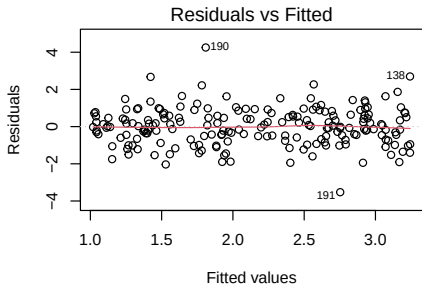
```
set.seed(647)
n = 200
X = runif(n)
u = rnorm(n)
Y = 1 + 2*X + u
fit <- lm(Y ~ X)

plot(X, Y)
abline(fit)

par(mfrow=c(2,2))
plot(fit)
```


Example: $E[X_i^4] < \infty$ and $E[Y_i^4] < \infty$





Example: Large Residual, Small Leverage

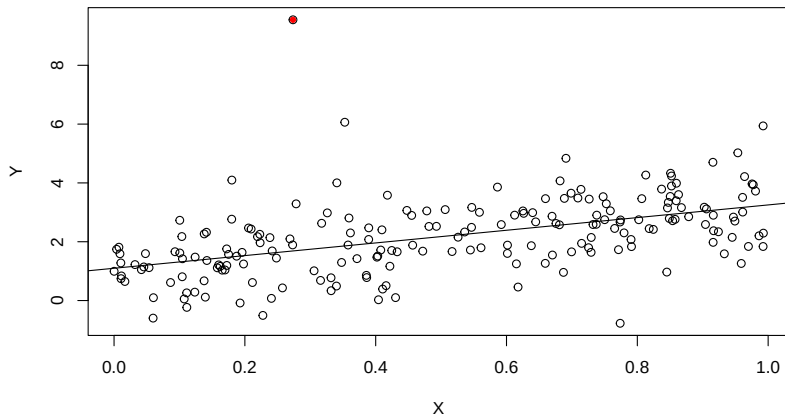
Example:

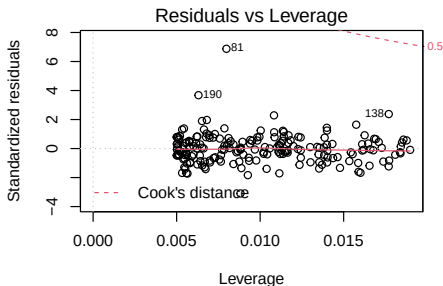
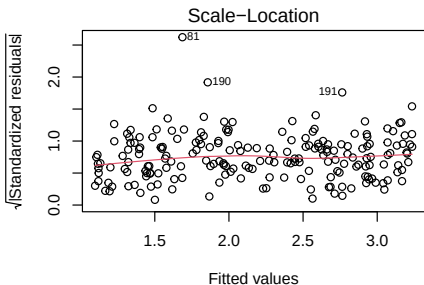
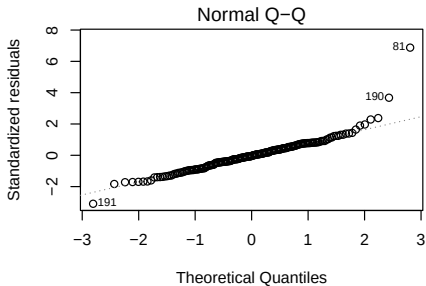
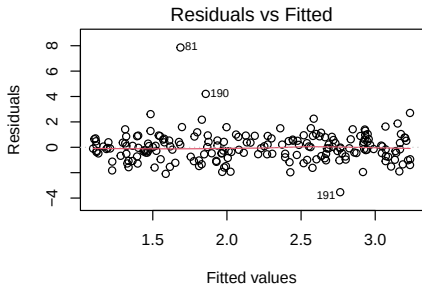
```
set.seed(647)
n = 200
X = runif(n)
u = rnorm(n)
Y = 1 + 2*X + u
Y[81] = 1 + 2*X[81] + 8
fit <- lm(Y ~ X)

plot(X, Y)
abline(fit)
lines(X[81], Y[81], pch = 20, type = "p", col = "red")

par(mfrow=c(2,2))
plot(fit)
```

Example: Large Residual, Small Leverage





Example: Large Residual, Large Leverage

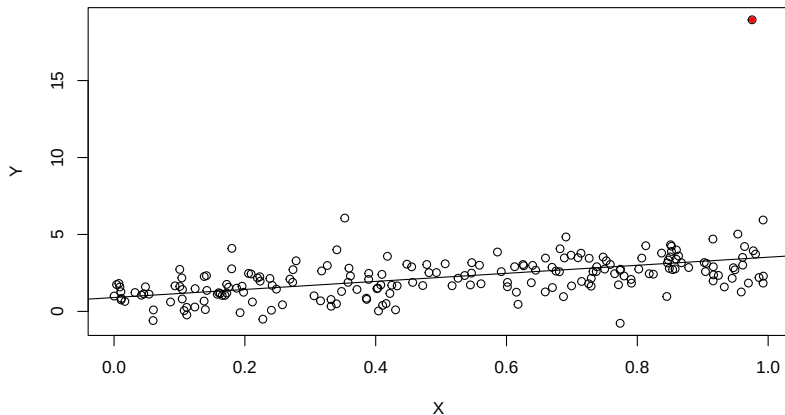
Example:

```
set.seed(647)
n = 200
X = runif(n)
u = rnorm(n)
Y = 1 + 2*X + u
Y[71] = 1 + 2*X[71] + 16
fit <- lm(Y ~ X)

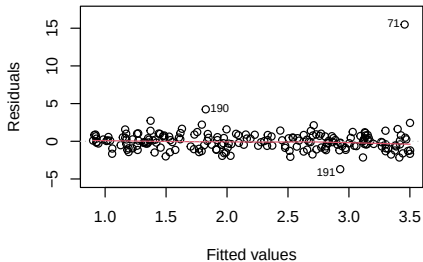
plot(X, Y)
abline(fit)
lines(X[71], Y[71], pch = 20, type = "p", col = "red")

par(mfrow=c(2,2))
plot(fit)
```

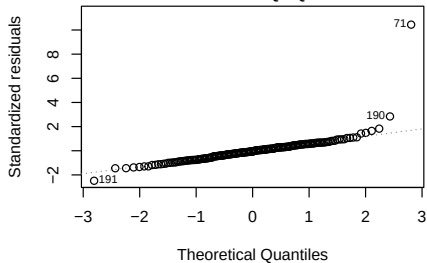
Example: Large Residual, Large Leverage



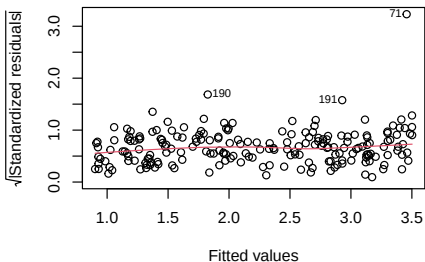
Residuals vs Fitted



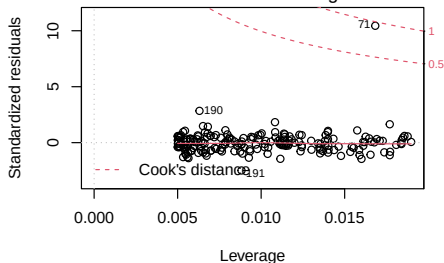
Normal Q-Q



Scale-Location



Residuals vs Leverage



Example: Small Residual, Large Leverage

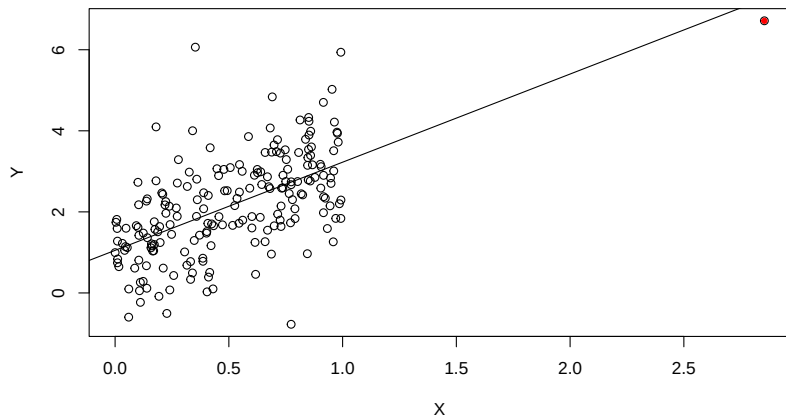
Example:

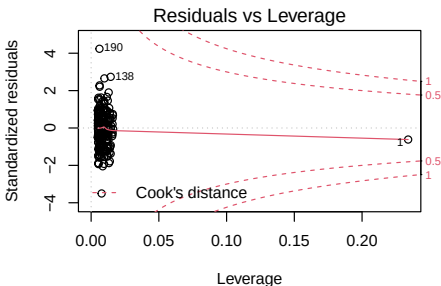
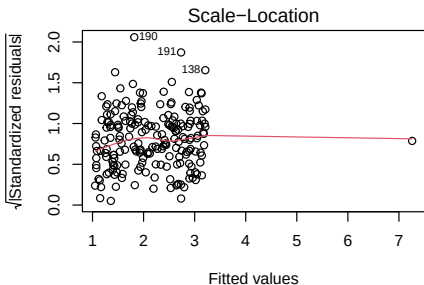
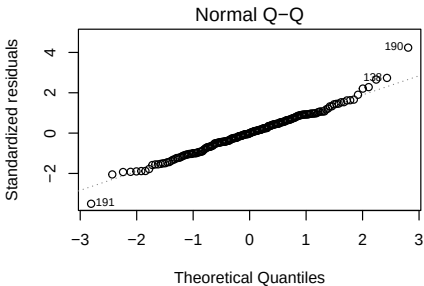
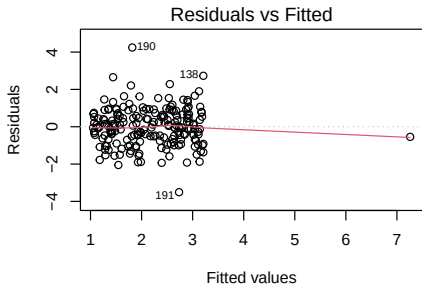
```
set.seed(647)
n = 200
X = runif(n)
u = rnorm(n)
Y = 1 + 2*X + u
X[1] = X[1] + 2
Y[1] = 1 + 2*X[1] + u[1]
fit <- lm(Y ~ X)

plot(X, Y)
abline(fit)
lines(X[1], Y[1], pch = 20, type = "p", col = "red")

par(mfrow=c(2,2))
plot(fit)
```

Example: Small Residual, Large Leverage





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Infeasible WLS: Known Heteroskedasticity

Assumptions:

We assume that there is the following relation between X_i and Y_i :

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

where, as always,

- ❶ *exogeneity* assumption: $E[u_i|X_i] = 0$
- ❷ (X_i, Y_i) is i.i.d. sample drawn from a joint distribution
- ❸ $E[X_i^4]$ and $E[Y_i^4]$ are nonzero and finite

and also assume

- ❹ $\text{Var}(u_i|X_i) = \lambda h(X_i)$, where λ is a constant and $h(\cdot)$ is a known function.

Infeasible WLS: Known Heteroskedasticity

So, for this model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with} \quad \text{Var}(u_i|X_i) = \lambda h(X_i),$$

the

- OLS estimator is unbiased and consistent
- errors can be obtained via *heteroskedasticity-robust standard error* estimator

What is the problem?

Infeasible WLS: Known Heteroskedasticity

So, for this model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with} \quad \text{Var}(u_i|X_i) = \lambda h(X_i),$$

the

- OLS estimator is unbiased and consistent
- errors can be obtained via *heteroskedasticity-robust standard error* estimator

What is the problem?

The problem is that the OLS estimator may be not the **B**est **L**inear conditionally **U**nbiased **E**stimator (i.e. not **BLUE**).

Infeasible WLS: Known Heteroskedasticity

We notice that this model,

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with} \quad \text{Var}(u_i|X_i) = \lambda h(X_i),$$

can be equivalently written as

$$\tilde{Y}_i = \beta_0 \tilde{X}_{i0} + \beta_1 \tilde{X}_{i1} + \tilde{u}_i,$$

where

$$\tilde{Y}_i = \frac{Y_i}{\sqrt{h(X_i)}}, \quad \tilde{X}_{i0} = \frac{1}{\sqrt{h(X_i)}}, \quad \tilde{X}_{i1} = \frac{X_i}{\sqrt{h(X_i)}} \quad \text{and} \quad \tilde{u}_i = \frac{u_i}{\sqrt{h(X_i)}}.$$

Def.

The OLS estimator of β_1 in the modified model $\tilde{Y}_i = \beta_0 \tilde{X}_{i0} + \beta_1 \tilde{X}_{i1} + \tilde{u}_i$ is called *weighted least squares* (WLS) estimator of β_1 in the original model for (X_i, Y_i) .

Infeasible WLS: Known Heteroskedasticity

Claim:

Given model

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with} \quad \text{Var}(u_i | X_i) = \lambda h(X_i),$$

the WLS estimator

- is **BLUE**;
- can be obtained by minimizing the weighted sum of squared residuals

$$\sum_{i=1}^n \frac{1}{h(X_i)} (Y_i - \beta_0 - \beta_1 X_i)^2.$$

Infeasible WLS: Known Heteroskedasticity

Claim:

Given model

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with} \quad \text{Var}(u_i | X_i) = \lambda h(X_i),$$

the WLS estimator

- is **BLUE**;
- can be obtained by minimizing the weighted sum of squared residuals

$$\sum_{i=1}^n \frac{1}{h(X_i)} (Y_i - \beta_0 - \beta_1 X_i)^2.$$

Note: Since one needs to know function $h(\cdot)$, which is rarely available, this estimator sometimes is called *infeasible* WLS.

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Feasible WLS: Heteroskedasticity of Known Function Form

Assumptions:

let's again assume that there is the following relation between X_i and Y_i :

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

where, as always,

- ❶ *exogeneity* assumption: $E[u_i|X_i] = 0$
- ❷ (X_i, Y_i) is i.i.d. sample drawn from a joint distribution
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and also assume

- ❹ $u_i^2 = h(X_i)\nu_i$ where $h(\cdot)$ is a function of some known form and $E[\nu_i|X_i] = 1$

Note:

In practice, to recover the form of $h(\cdot)$ one needs to plot the estimated (via OLS) errors $\hat{u}_i^2$ vs. X_i .

Feasible WLS: Heteroskedasticity of Known Function Form

Claim:

For this model

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } u_i^2 = h(X_i)\nu_i, \quad \text{where } E[\nu_i|X_i] = 1,$$

the conditional variance of errors is

$$\text{Var}(u_i|X_i) = h(X_i).$$

Feasible WLS: Heteroskedasticity of Known Function Form

Claim:

For this model

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad \text{with } u_i^2 = h(X_i)\nu_i, \quad \text{where } E[\nu_i|X_i] = 1,$$

the conditional variance of errors is

$$\text{Var}(u_i|X_i) = h(X_i).$$

Corollary:

The dependence $h(X_i)$ of $\text{Var}(u_i|X_i)$ on X_i can be estimated via regression of u_i^2 on X_i and/or its transformations.

For example,

- if u_i^2 depends on X_i linearly, i.e. $h(X_i) = a + bX_i$, one can regress u_i^2 on X_i to get $\hat{h}(\cdot)$;
- if u_i^2 depends on X_i quadratically, i.e. $h(X_i) = a + bX_i^2$, one can regress u_i^2 on X_i^2 to get $\hat{h}(\cdot)$.

The WLS estimator, together with $\hat{h}(\cdot)$, then can be used to estimate β_1 . This estimator is sometimes called *feasible* WLS.

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Example: Feasible WLS Estimator

Example:

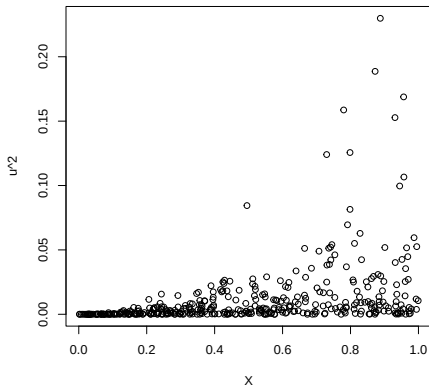
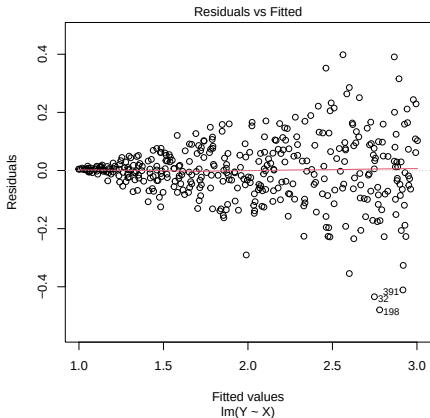
```
set.seed(543)
n = 400
X = runif(n, min = 0, max = 1)
Y = 1 + 2*X + X * rnorm(n, sd = 0.2)

fit_OLS <- lm(Y ~ X)
u = fit_OLS$residuals

plot(fit_OLS, which = 1)
plot(X, u^2)
```


Example: Feasible WLS Estimator

Example: (continued)



Example: Feasible WLS Estimator

Example: (continued)

```
fit_h <- lm(I(u^2) ~ I(X^2))
```

```
hi <- predict(fit_h)
```

```
fit_OLS_robust <- coeftest(fit_OLS, vcov = vcovHC, type = "HC1")
```

```
fit_WLS <- lm(I(Y/sqrt(hi)) ~ -1 + I(1/sqrt(hi)) + I(X/sqrt(hi)))
```

```
stargazer(fit_OLS, fit_OLS_robust, fit_WLS,  
          type = "text",  
          title="(1) Homoskedastic OLS;  
                (2) Heteroskedastic OLS;  
                (3) WLS")
```

Example: Feasible WLS Estimator

(1) Homoskedastic OLS; (2) Heteroskedastic OLS; (3) WLS

	Dependent variable:		
	Y		I (Y/sqrt(hi))
	OLS	coefficient test	OLS
	(1)	(2)	(3)
X	2.010*** (0.019)	2.010*** (0.020)	
I(1/sqrt(hi))			0.996*** (0.004)
I(X/sqrt(hi))			2.005*** (0.014)
Constant	0.995*** (0.011)	0.995*** (0.007)	
Observations	400		400
R2	0.967		0.999
Adjusted R2	0.967		0.999
Residual Std. Error (df = 398)	0.108		0.943
F Statistic	11,739.070*** (df = 1; 398)		135,383.800*** (df = 2; 398)

Note: *p<0.1; **p<0.05; ***p<0.01

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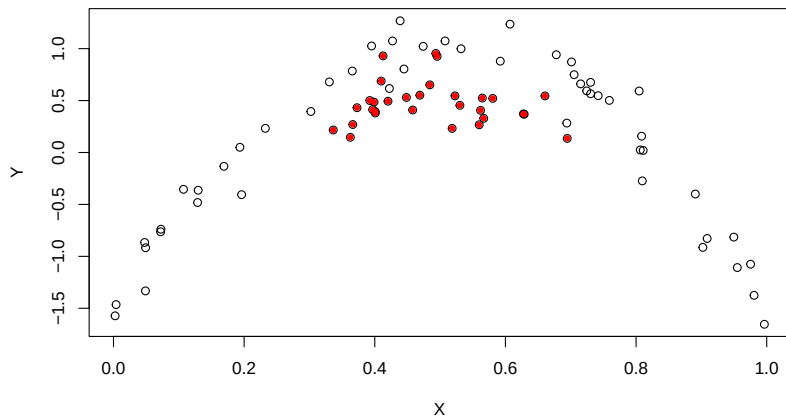
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Matching Methods: Example

```
set.seed(543)
nC = 50
nT = 30
n = nC+nT
XC = runif(nC)
XT = runif(nT, 0.3, 0.7)
X = c(XC,XT)
D = c(rep(0,nC),rep(1,nT))
u = rnorm(n,sd=0.2)
Y = 1 -10*(X-0.5)^2 - 0.4*D + u
plot(X,Y)
lines(X[(n-nT+1):n], Y[(n-nT+1):n],
      pch = 20, type = "p", col = "red")
fit_lin <- lm(Y ~ D + X)
fit_quad <- lm(Y ~ D + X + I(X^2))
stargazer(fit_lin, fit_quad,
          type = "text",
          title="(1) Linear; (2) Quadratic")
```

Matching Methods: Example



Matching Methods: Example

Example:

(1) Linear; (2) Quadratic

=====		
	Dependent variable:	

	Y	
	(1)	(2)

D	0.449*** (0.159)	-0.481*** (0.054)
X	0.271 (0.302)	10.368*** (0.343)
I(X2)		-10.293*** (0.339)
Constant	-0.112 (0.185)	-1.563*** (0.070)

Observations	80	80
R2	0.099	0.931
Adjusted R2	0.076	0.929
Residual Std. Error	0.688 (df = 77)	0.191 (df = 76)
F Statistic	4.226** (df = 2; 77)	343.831*** (df = 3; 76)
=====		

Note: *p<0.1; **p<0.05; ***p<0.01

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Matching Methods: Exact Matching

Exact matching (or *cell matching*) is defined as follows:

- ▶ Assume X_i takes K distinct values
- ▶ Let D_i be 1 if obs. i is treated and 0 otherwise
- ▶ Let N_{1k} and N_{0k} denote the number of treated and untreated observations in cell $k \in \{1, 2, \dots, K\}$
- ▶ Let $C(k)$ denotes the set of indexes i from the same cell
- ▶ The *average treated effect on treated* is estimated as

$$\hat{\Delta}^{ATT} = \sum_{k=1}^K \frac{N_{1k}}{N_1} \left(\sum_{i \in C(k) \cap \{D_i=1\}} \frac{y_i}{N_{1k}} - \sum_{i \in C(k) \cap \{D_i=0\}} \frac{y_i}{N_{0k}} \right)$$

- ▶ The *average treated effect on untreated* is estimated as

$$\hat{\Delta}^{ATU} = \sum_{k=1}^K \frac{N_{0k}}{N_0} \left(\sum_{i \in C(k) \cap \{D_i=1\}} \frac{y_i}{N_{1k}} - \sum_{i \in C(k) \cap \{D_i=0\}} \frac{y_i}{N_{0k}} \right)$$

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- ▶ The *average treated effect* is then estimated as

$$\hat{\Delta}^{ATE} = \frac{N_1}{N} \hat{\Delta}^{ATT} + \frac{N_0}{N} \hat{\Delta}^{ATU}$$